HDOC Day 63

Problem Statement

Write a C program to read the size and elements of a variable-length integer array containing distinct numbers from 0 to size, with exactly one number missing. Pass the array address to a function named missing_number() using pointers. Find and display the missing integer.

C Program

```
#include <stdio.h>

void missing_number(int *nums, int size) {
    int expected = size * (size + 1) / 2;
    int actual = 0;

    for (int i = 0; i < size; i++) {
        actual += *(nums + i);
    }

    printf("Missing number: %d\n", expected - actual);
}

int main() {
    int size;
    printf("Enter the size of array: ");
    scanf("%d", &size);

    if (size <= 0) {
        printf("Size must be positive.\n");
        return 1;
    }

    int nums[size];
    printf("Enter %d distinct numbers from 0 to %d:\n",
           size, size);
    for (int i = 0; i < size; i++) {
        scanf("%d", &nums[i]);
    }

    missing_number(nums, size);
    return 0;
}
```

Sample Input and Output

```
Enter the size of array: 5
Enter 5 distinct numbers from 0 to 5:
0 1 3 4 5
Missing number: 2
```

Explanation

The sum of all integers from 0 to size is size * (size + 1) / 2. The function adds the array elements using pointer arithmetic (*(nums + i)). The difference between the expected sum and actual sum gives the

missing number. The array is passed by its address.